Diagnosis: Non-small cell lung cancer

Treatement 1: Afatinib

Chosen clinical trials:

1. 'Understanding Mechanisms of Acquired Resistance to BIBW2992' completed bibw 2992 https://clinicaltrials.gov/study/NCT01074177

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

1. 'GIOTRIF rPMS in Korean Patients With NSCLC' completed giotrif 20mg giotrif 40mg giotrif 30mg https://clinicaltrials.gov/study/NCT02285361

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

1. 'LUX-Lung 4: BIBW 2992 (Afatinib) Phase I Trial in Advanced Non Small Cell Lung Cancer Patients & Phase II Trial in Non Small Cell Lung Cancer Patients Failing Erlotinib or Gefitinib' completed bibw 2992 ma2 40mg/day bibw 2992 ma2 50mg/day bibw 2992 ma2 20mg/day bibw 2992 qd https://clinicaltrials.gov/study/NCT00711594

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

1. 'The Study Observes How Long Patients With Non-small Cell Lung Cancer (NSCLC) Benefit From Treatment With Epidermal Growth Factor Tyrosine Kinase Inhibitor (EGFR-TKI) When Given Either for Uncommon Mutations or for Common Mutations in the Sequence Afatinib Followed by Osimertinib' completed afatinib (gi(l)otrif®) erlotinib (tarceva®) gefitinib (iressa®) osimertinib (tagrisso®) afatinib (gi(l)otrif®) osimertinib (tagrisso®) https://clinicaltrials.gov/study/NCT04179890

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

1. 'Evaluate the Impact of Afatinib on Quality of Life and Symptom Burden of Greek Subjects With Advanced NSCLC in Routine Patient Care Settings' completed afatinib https://clinicaltrials.gov/study/NCT02440854

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

1. 'LUX-Lung 5: Afatinib Plus Weekly Paclitaxel Versus Investigator's Choice of Single Agent Chemotherapy Following Afatinib Monotherapy in Non-small Cell Lung Cancer Patients Failing Erlotinib or Gefitinib' completed investigator´s choice of chemotherapy bibw 2992 https://clinicaltrials.gov/study/NCT01085136

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

1. 'Pembrolizumab and Afatinib in Patients With Non-small Cell Lung Cancer With Resistance to Erlotinib' completed afatinib dimaleate https://clinicaltrials.gov/study/NCT02364609

   Population: 14

   Time: At the time of disease progression (median duration of 11.4 months from start of treatment)

   T790M Mutation: 4

Chosen scientific papers:

Afatinib is approved for first-line treatment in advanced EGFR mutation-positive NSCLC [1]. Afatinib showed potential in improving treatment outcomes for NSCLC patients [2]. Afatinib showed efficacy in older adult EGFR-mutant lung cancer patients [3].

[1] Prospective Observational Real-World Study of Afatinib as First-Line Treatment in Patients with Epidermal Growth Factor Receptor (EGFR) Mutation-Positive Advanced Non-Small Cell Lung Cancer (NSCLC) in China. https://pubmed.ncbi.nlm.nih.gov/41736995

[2] EGFR and IRE1α pathways are associated with distinct immunomodulatory gene expression profiles in NSCLC cells with acquired resistance to EGFR TKIs. https://pubmed.ncbi.nlm.nih.gov/41672191

[3] Efficacy and safety findings of the EXTRA study in older adult EGFR-mutant lung cancer patients receiving afatinib as first-line treatment. https://pubmed.ncbi.nlm.nih.gov/41673455